RoadBlockagePlugin.py User Manual

Program Overview

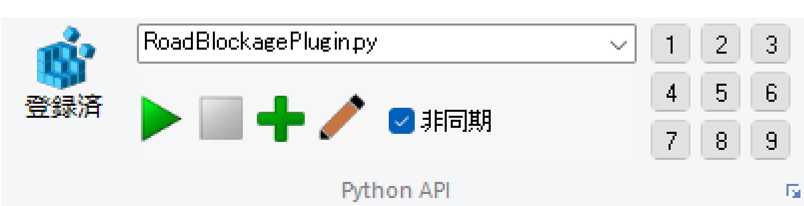
`RoadBlockagePlugin.py` is a Python plugin that runs inside UC-win/Road. It loads a CSV file (a grid of latitude, longitude, and probability) that predicts the probability of road blockage occurring due to events such as earthquakes, and automatically places a **'Barricade'** 3D model as a visual marker on the nearest lane at every location where the probability exceeds a threshold.

Because the UC-win/Road SDK provides no way to newly create an actual road obstruction in UC-win/Road, the Barricade is only a visual marker indicating "an obstruction should be placed here." Using this marker as a guide, the user manually places the actual road obstruction in UC-win/Road.

The plugin also includes a **Traffic Monitor** feature that lets the user select any single point and measure and graph the surrounding traffic flow (average speed and vehicle count) in real time. By watching this monitor, the user can check how vehicles (ambient traffic) actually react to a real road obstruction — that is, how their speed and count change.

How to Run the Program

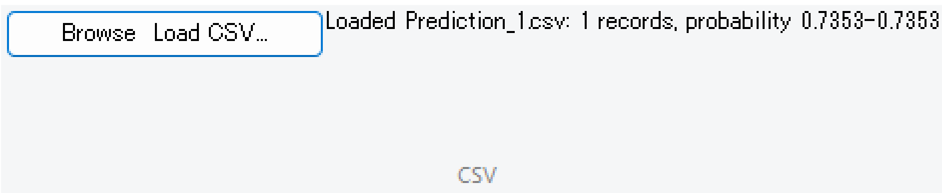
1. Open the project in UC-win/Road.
2. Put the simulation into Play state (Barricades will not be placed correctly unless the simulation is playing).
3. Load the plugin `RoadBlockagePlugin.py` in UC-win/Road (enable the Async option when doing so).
4. Once running, the following five tabs are added to the UC-win/Road ribbon:
    - **CSV**: Load a CSV file
    - **Barricades**: Place Barricades
    - **Traffic Monitor**: Display the traffic flow monitor
    - **Find Objects**: Jump to an object
    - **Reset**: Reset



Operation Flow

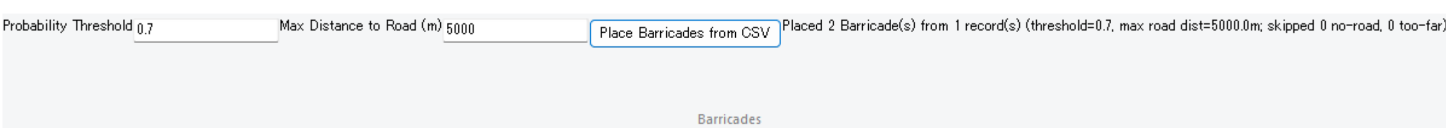
Loading the CSV generated by RoadBlockDetectionApp.py

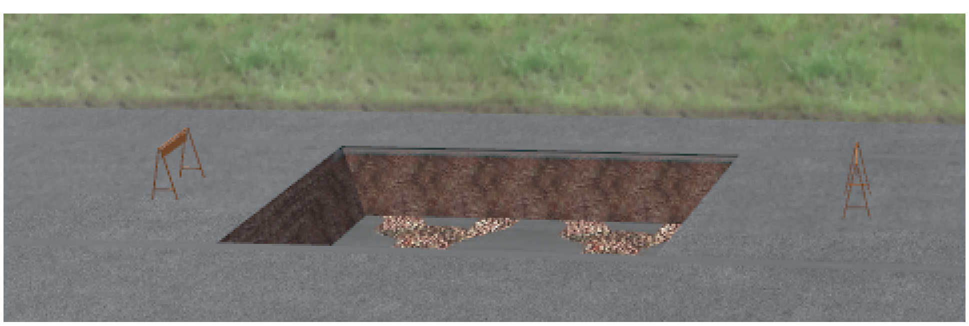
1. In the **CSV** tab, click **"Browse & Load CSV..."**.
2. Select a CSV file (sample outputs are provided in the `Prediction/` folder).
3. On successful loading, a summary of the CSV file is shown in the label.



Placing Barricades

1. In the **Barricades** tab, configure the following:
   - **Probability Threshold**: only CSV records whose probability is at or above this value are placed.
   - **Max Distance to Road (m)**: records whose distance to the nearest road exceeds this value are treated as "not on a road" and skipped.
2. Click **"Place Barricades from CSV"**. This finds the road and lane nearest to each record's latitude/longitude, then places Barricades **two per record**, offset **5 m upstream and 5 m downstream** along the road from that center point.
3. The user manually places the real road obstruction using the midpoint between these two Barricades as a guide.





Displaying the Monitor (visualizing average speed)

1. In the **Traffic Monitor** tab, configure the following:

- **CSV Record Index**: the index of the CSV record to monitor.
- **Measurement Radius (m)**: vehicles within this radius are measured.

2. Click **"Start Monitoring"**.
   - The nearest position on a road to the latitude/longitude of the specified CSV record is found and used as the monitoring reference point.
3. The HUD window and the ribbon's detail field display the following in real time at 0.5-second intervals:
   - **Avg Speed (km/h)**: the average speed of vehicles within the radius, plus a line graph of its trend.
   - **Vehicles in Zone**: the current vehicle count within the radius, plus a line graph of its trend.